Remote Sensing Hyperspectral Image Super-Resolution via Multidomain Spatial Information and Multiscale Spectral Information Fusion

Chi Chen, Yongcheng Wang, *Member, IEEE*, Yuxi Zhang, Zhikang Zhao, and Hao Feng

***Abstract*— Hyperspectral image (HSI) super-resolution (SR) technology has made remarkable progress due to the development of deep learning (DL). However, the technique still faces two challenges, i.e., the imbalance between spectral and spatial information extraction, and the parameter deviation and high computational effort associated with 3-D convolution. In this article, we propose an SR method for remote sensing HSIs based on multidomain spatial information and multiscale spectral information fusion (MSSR). Specifically, inspired by the high degree of self-similarity of remote sensing HSIs, a spatial–spectral attention module based on dilated convolution (DSSA) for capturing global spatial information is proposed. The extraction of local spatial information is then accomplished by residual blocks using small-size convolution kernels. Meanwhile, we propose the 3-D inception module to efficiently mine multiscale spectral information. The module only retains the scale of the 3-D convolution kernel in spectral dimension, which greatly reduces the high computational cost caused by 3-D convolution. Comparative experimental results on four benchmark datasets demonstrate that compared with the current cutting-edge models, our method achieves state-of-the-art (SOTA) results and the model computation is greatly reduced.**

***Index Terms*— 3-D inception, dilated convolution, hyperspectral image (HSI), spatial–spectral attention, super-resolution (SR).**

I. Introduction

HYPERSPECTRAL images (HSIs) have dozens or even hundreds of spectral bands. The rich spectral information reflects the spectral characteristics of different materials, which is conducive to more accurate detection [1], [2], [3] and classification [4], [5], [6]. Due to the mutual constraints between spectral resolution and spatial resolution, HSIs are usually accompanied by low spatial resolution. It makes the spectral interpretation work "rough," and the purpose of super-resolution (SR) is to make the image refined. Unlike hardware upgrading, image SR reconstruction has great application prospects and value as a low-cost and high-efficiency image postprocessing technique.

Early work on HSI SR was dominated by traditional methods. The mainstream technical routes can be categorized into three types, which are based on wavelet transform [7], [8], [9], maximum a posteriori estimation [10], [11], [12], and spectral mixing analysis [13], [14], [15], [16], [17], of which the representative works include the CNMF algorithm [13]. Under normal circumstances, the disadvantage of traditional methods is that the solution is difficult and time-consuming. Dong et al. [18] introduced a convolutional neural network (CNN) into the image SR task for the first time, realizing the leap from traditional algorithms to deep learning (DL). Although the development of natural image SR technology based on DL has entered a relatively mature stage [19], [20], [21], [22], the methods for HSIs are still very limited. At present, there are two kinds of HSI SR methods, which are multiimage fusion and single-image SR (SISR). Multiimage fusion can be further subdivided into fusion based on multispectral image (MSI) and panchromatic (PAN) sharpening. MSI usually has lower spectral resolution and higher spatial resolution than HSI. Palsson et al. [23] combined principal component analysis (PCA) with 3-D convolutional networks to deal with MSI-HSI fusion problem as early as in 2017. Inspired by the deep image prior, Zhang et al. [24] realized blind SR for HSIs by co-training the generator and degrader, which greatly extends the application scenarios of this technique. PAN images generally have superior spatial resolution. Qu et al. [25] constructed a multipath network for feature extraction on different scales for target HSIs and auxiliary PANs. Meanwhile, using multiple shallow networks for feature extraction can effectively reduce the computational cost. The image segmentation-based injection gain estimation algorithm in GASN [26] can achieve a relative balance between the oversharpening problem and the fuzzy effect. Unlike CNNs that focus more on local dependencies, Meng et al. [27] proposed a ViT-based pansharpening network to comprehensively utilize global information. Bandara and Patel [28] proposed hyper-transformer, which utilizes multi-head feature soft-attention network to extract the cross-feature space dependencies of PAN and LR-HSI. Zhou et al. [29] introduced the knowledge-distilling strategy into the field of pansharpening for the first time, which effectively enhances the knowledge transfer and information fusion between heterogeneous features. Image fusion can complement external prior information and usually gets excellent reconstruction

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results. However, the premise of image fusion is to obtain auxiliary MSI or PAN accurately registered with the target HSI [30], which greatly limits the application range of the method. Therefore, SISR seems to be a more widely applied direction. The 3D-FCNN [31] and GDRRN [32] are both SISR models with relatively simple structures. In addition, SSPSR [33] and GELIN [34] are both methods dealing with HSI SR tasks based on band grouping, while MCNet [35] and ERCSR [36] are representative works exploring 3-D convolution.

The value of HSIs is mainly reflected in the rich spectral information. In order to ensure the accuracy of subsequent spectral interpretation [37], it is necessary to improve the spectral fidelity of the reconstructed image as much as possible. Most of the works at this stage overemphasize the extraction of spectral information, but the extraction of spatial information is not sufficient. Previous studies have demonstrated that there is an obvious self-similarity inside the image [38], [39], [40], and this property reminds us that global information can provide rich prior knowledge for local refinement of texture. Small-size patches usually recur in the same image. Since remote sensing HSIs usually have very low spatial resolution, this recurrence phenomenon is particularly obvious in the observed scene. Therefore, we propose a spatial–spectral attention mechanism based on dilated convolution (DSSA) for extracting global spatial information. Dilated convolution makes it possible to obtain a larger receptive field by using only the shallow network, and can effectively mine the internal prior information with more predictive power than the external prior. In addition, we use residual blocks with a convolution kernel size of 3×3 for local spatial information extraction [41], [42].

The sliding of the convolution kernel on spectral dimension makes 3-D convolution more suitable for exploring spectral information than 2-D convolution. However, there are two problems with the use of 3-D convolution. One is the high computational cost, and the other is the parameter deviation that results when the convolution kernel simultaneously fits the feature distributions in three dimensions. To address the above problems, in this article, we construct a 3-D inception module. Specifically, we use a convolution kernel in the shape of $k \times 1 \times 1$ to replace the conventional $k \times k \times k$ convolution kernel, and different sizes of k are chosen for different branches in the module for mining spectral correlations at different scales. Since only the repeated calculation process of convolution in spectral dimension is preserved, the computational effort is greatly reduced and is able to take on the task of specializing in spectral information.

Overall, the contributions of this article are as follows.

1) A three-branch network MSSR is constructed, and each branch is responsible for global spatial information extraction, local spatial information extraction, and spectral information extraction, respectively. The reconstructed image derived from the fusion feature of the three has both rich spatial information and realistic spectral information.
2) A new spatial–spectral attention mechanism DSSA is proposed. The module realizes the extraction and integration of global spatial information across scales and effectively alleviates the problem of inadequate spatial information extraction.
3) To efficiently mine multiscale spectral information of images, we propose a 3-D inception module. In this module, all convolution kernels only retain the scale in spectral dimension, which greatly reduces the high computational amount introduced by 3-D convolution.
4) Comparative experiments on four benchmark datasets show that our method achieves state-of-the-art (SOTA) results in peak signal-to-noise ratio (PSNR), structural similarity (SSIM) [43], spectral angle mapper (SAM) [44], and erreur relative globale adimensionnelle de synthèse (ERGAS) [45] with a several-fold reduction in modeling computation compared to the cutting-edge models.

The rest of the article is organized as follows. In Section II, we introduce some classical SR methods for single HSI and some works related to 3-D convolution. Section III describes our proposed MSSR model in detail, including the overall architecture, the DSSA module, and the 3-D inception module. In Section IV, we report the results of the comparison experiments between the proposed method and other SOTA methods. Meanwhile, ablation experiments are carried out for the DSSA and 3-D inception modules to validate the effectiveness of our method. Finally, the conclusion is given in Section V.

II. Related Works

A. Single HSI SR

Single HSI SR can get rid of the limitation of registered data and effectively reduce the industrial cost, which has high application value and practical significance. Next, we will briefly review the development of classical single HSI SR.

Some of the earlier work can be seen as an extension of natural image SR methods, among which the classics include 3D-FCNN proposed by Mei et al. [31] and GDRRN proposed by Li et al. [32]. The former is a simple four-layer convolutional network. The authors use 3-D convolution instead of the common 2-D convolution, aiming at mining spectral correlations among different bands. The latter can be regarded as the transformation of the DRRN [46] model for HSI, with the difference that the grouped convolution is introduced to reduce the number of model parameters. Inspired by transfer learning, Yuan et al. [47] proposed to first learn the mapping relationship between high-resolution (HR)–low-resolution (LR) natural image pairs through neural networks, and then transfer such mapping relationship to the HSI domain. It proved that there is a certain commonality between different image domains. Li et al. [48] proposed a band-attention mechanism that can be embedded into the generator. It aims to improve the spectral distortion problem, but the whole model pays relatively limited attention to spatial information. Since HSIs usually have hundreds of bands, this data complexity makes the model very easy to overfit and computationally intensive. Therefore, Jiang et al. [33] proposed to split the complete image from spectral dimension into several subimages with a small number of spectral bands. The feature of each subimage is extracted and then fused into a complete HSI. It is worth noting that the model adopts the progressive upsampling framework, and the upsampling

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operation occurs separately in each branch and the final fusion process. Also based on the band grouping strategy, in order to better utilize the complementary information between different bands, Wang et al. [34] proposed the neighboring group integration module. In addition, the model reconstructs each subimage individually, which can effectively reduce the high computational effort caused by high-dimensional data. Liu et al. [49] proposed a two-branch network to extract spatial and spectral information, respectively, which is the first time that a transformer is used to solve the task of a single HSI SR. This special structure stacked by the attention mechanism can well explore global information. Unlike most of the methods that process spatial and spectral information separately, Hu et al. [50] integrated the complete information extraction and utilization process in an integrated network, providing a new idea for HSI SR. Different HSIs often have different numbers of spectral bands, and this raw data limitation makes it difficult to construct a unified neural network entrance. To solve this problem, Gong et al. [51] designed an input processing module to adjust the structure of the input images and overcome the limitation of network selection.

Nowadays, more and more scholars are focusing on single HSI SR methods. With the advancement of industrial technology, the current problem of dataset shortage will be solved eventually. When sufficient external prior information is obtained, the performance of a single HSI SR method will reach a new height and usher in better development.

B. Three-Dimensional Convolution

The greatest advantage of HSI lies in the ability to capture the spectral signal of the observed scene, which is also an important support for subsequent spectral interpretation. In the pursuit of spectral fidelity of reconstructed images, some scholars found that 3-D convolution performs well in mining spectral correlation compared to the commonly used 2-D convolution.

Three-dimensional-FCNN [31] is a pioneer work in this direction. Jiang et al. [52] used 3-D convolution in combination with generative adversarial network (GAN), which is the first time GAN has been used for HSI SR tasks. Such conventional 3-D convolution usually brings overwhelming computational costs. Scholars began to seek some improvements to the use of 3-D convolution, among which the representative works are MCNet [35] and ERCSR [36] models. MCNet is proposed to solve the problem of unbalanced extraction of spatial–spectral information. Specifically, a 2-D/3-D mixed convolutional module is proposed to better mine potential spatial information. In addition, the separable 3-D convolution in the 3-D units can effectively reduce the computational effort compared to the conventional 3-D convolution. However, the parallel structure in this network leads to the inevitable module redundancy problem. Afterward, the ERCSR proposed by the same team mitigates the problem by alternately connecting 2-D and 3-D units. In 3-D units, the network is allowed to mine information in all directions synchronously by splitting the convolutional kernel of $k \times k \times k$ into $1 \times k \times k$, $k \times 1 \times k$, and $k \times k \times 1$.

Although scholars have already put forward various ideas, the problem of the high computational effort of 3-D convolution has not been well solved. The 3-D inception module proposed in this article aims to specialize the utility of 3-D convolution and effectively reduces the computational amount by retaining only the repeated calculations in spectral dimension.

III. Proposed Method

In this section, we introduce the proposed SR model MSSR for remote sensing HSIs in detail. As shown in Fig. 1, the MSSR is a three-branch network, which consists of the global spatial information extraction branch, the local spatial information extraction branch, and the spectral information extraction branch. The global spatial information extraction branch is recursively connected by the DSSA module. The local spatial information extraction is accomplished by using residual blocks based on small-size convolution kernels. And 3-D inception module is used for multiscale spectral information extraction. Next, we will first introduce the overall framework of the model, and then elaborate on the specific construction of DSSA and 3-D inception, respectively.

A. Overall Framework

It is defined that $I_{\mathrm{HR}} \in \mathbb{R}^{sH \times sW \times C}$ represents the ground truth image with HR, while the corresponding LR image is denoted by $I_{\mathrm{LR}} \in \mathbb{R}^{H \times W \times C}$, where s represents the scaling factor. We named the proposed SR model MSSR, which aims at recovering the I_{SR} as close as possible to the ground truth image from the LR HSI, i.e.,

$$I_{\mathrm{SR}} = F_{\mathrm{MSSR}}(I_{\mathrm{LR}}). \tag{1}$$

The next step is to sort out the overall network.

First, the input image I_{LR} passes through a primary feature extraction module consisting of three ordinary convolutional layers, denoted by f_{primary}, whose output is

$$I_{\mathrm{primary}} = f_{\mathrm{primary}}(I_{\mathrm{LR}}). \tag{2}$$

Then, the obtained feature map will be fed into three branches. The first branch is the global spatial information extraction branch f_{G_Spatial}, whose main body consists of DSSA blocks. In order to reduce the number of parameters, each module shares parameters, i.e., it uses each block recursively. The process can be formulated as

$$I_{G_\mathrm{Spatial}} = f_{G_\mathrm{Spatial}}\left(I_{\mathrm{primary}}\right) = f^n f^{n-1}, \ldots, f^2 f^1\left(I_{\mathrm{primary}}\right) \tag{3}$$

where n represents the number of recursions.

The second branch is the local spatial information extraction branch f_{L_Spatial}, whose body consists of cascaded residual blocks. It is worth noting that each layer in the residual block uses small convolutional kernels of size 3×3 to better mine the local information. Its output is

$$I_{L_\mathrm{Spatial}} = f_{L_\mathrm{Spatial}}\left(I_{\mathrm{primary}}\right). \tag{4}$$

The third branch is the multiscale spectral information extraction branch f_{Spectral}, which is accomplished by the proposed 3-D inception module, and the details of this module will be provided later. The output of this branch is

$$I_{\mathrm{Spectral}} = f_{\mathrm{Spectral}}\left(I_{\mathrm{primary}}\right). \tag{5}$$

After that, the outputs of the three branches are summed and fed into the reconstruction module. In this module, transposed convolution is chosen to perform the upsampling

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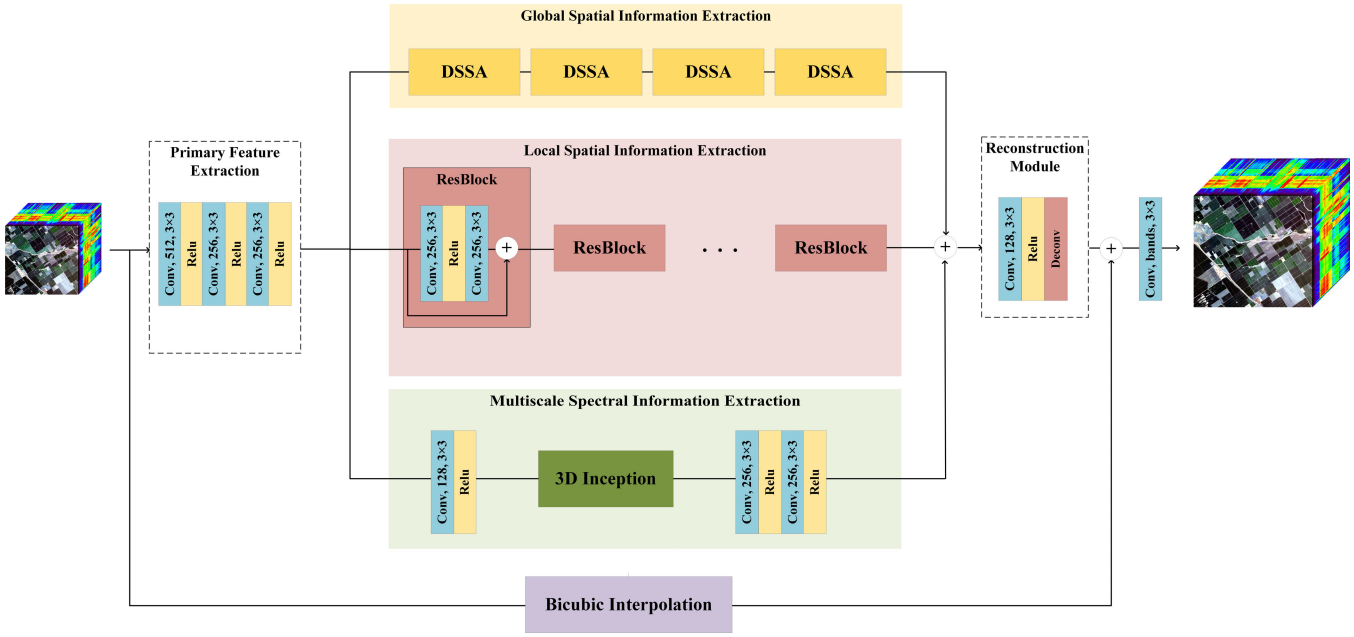


Fig. 1. Structure of the proposed MSSR, which consists of the global spatial information extraction branch, the local spatial information extraction branch, and the spectral information extraction branch.

operation. It allows the feature extraction to take place in the low-dimensional space, which can effectively reduce the computation amount. The process can be represented as

$$I_{\text{Reconstruct}} = f_{\text{Reconstruct}}\left(I_{G_\text{Spatial}} + I_{L_\text{Spatial}} + I_{\text{Spectral}}\right). \quad (6)$$

The final output is provided by summing the reconstructed image $I_{\text{Reconstruct}}$ with the upsampled image obtained by bicubic interpolation of I_{LR}, i.e.,

$$I_{\text{SR}} = f_{\text{Conv}}(I_{\text{Reconstruct}} + \text{BIC}(I_{\text{LR}})) \quad (7)$$

where f_{Conv} represents the last convolution layer after the two are added together for better fusion.

B. Spatial–Spectral Attention Mechanism Based on Dilated Convolution

HSIs have special requirements for spectral information. In order to alleviate the problem of spectral distortion, many scholars focus their research on the mining and integration of spectral correlation. This can easily lead to a decrease in attention to spatial information. Remote sensing HSIs have obvious self-similarity, and small-sized patches have a tendency to recur at different scales in different regions within the same image. The rich internal prior possesses a higher confidence than the external prior, which can provide strong support for reconstructing the refined texture of the image. Therefore, how to realize global spatial information extraction and cross-domain feature supplementation is particularly important in image restoration tasks. Based on the above demands, we propose a spatial–spectral attention module, whose structure is shown in Fig. 2.

To capture global information through conventional convolution, the feasible methods include deepening the network or increasing the size of the convolution kernel, but both options lead to a larger number of parameters and higher training difficulty. In order to better extract global information, we introduce dilated convolution [53] into the attention module. It allows a larger receptive field to be obtained over a relatively shallow network, largely preserving the internal data structure. In dilated convolution, i.e., adding voids to regular convolution, the hyperparameter dilation rate determines the number of added voids. Different dilation rates will bring different sizes of receptive fields, and the ability to obtain cross-scale information is also not available in conventional convolution. Specifically, we first feed the feature map into dilated convolution branches of different scales, i.e.,

$$I_3^1 = f_{d,3}\left(I^0\right) \quad (8)$$

where 1 in I_3^1 represents the order of the attention recursive blocks, 3 represents the size of the dilated convolution kernel, and I^0 represents the features fed into the DSSA module. Then, the global information extraction module can be expressed as

$$I_d^1 = I_3^1 + I_5^1 + I^0. \quad (9)$$

The extracted feature information I_d^1 will be fed into the spatial weight branch and the spectral weight branch.

Conventional spatial attention mechanisms directly process feature maps by average or max pooling, which is too "rude" for image restoration tasks such as SR. This kind of violent dimensionality reduction will lose the dimensional information of the feature map. Once the dimensional information is lost, it is extremely difficult to get it back from the existing information. Especially for 3-D data cubes such as HSI, directly compressing the spectral dimension into a single channel can easily lead to spectral distortion. Therefore, we use two 1×1 convolutional layers to complete the stepwise dimensionality reduction of the spectral bands. The 1×1 convolutional layer has excellent channel interaction capability and retains the spectral correlation information while reducing the dimensionality. This part can be expressed as

$$I_{\text{Spatial}}^1 = f_{\text{Spatial}}\left(I_d^1\right) * I^0 \quad (10)$$

where f_{Spatial} represents the function that finds the global spatial weights.

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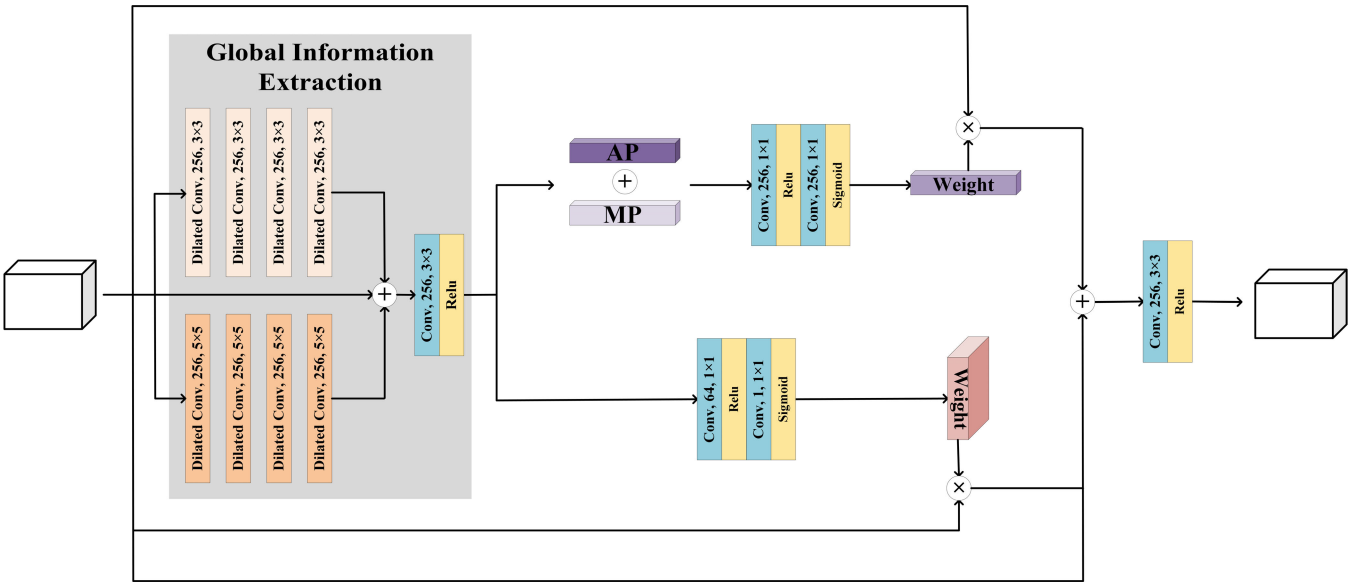


Fig. 2. Structure of the proposed DSSA. AP and MP represent max pooling and average pooling, respectively.

As for the spectral weight branch, we also use 1×1 convolution instead of the commonly used fully connected layer. This stage can be expressed as

$$I^1_{\text{Spectral}} = f_{\text{Spectral}}\left(I^1_d\right) * I^0 \tag{11}$$

where f_{Spectral} represents the function of obtaining the spectral weight.

The final output of the DSSA module is

$$I^1 = I^1_{\text{Spatial}} + I^1_{\text{Spectral}} + I^0. \tag{12}$$

In MSSR, we use the module recursively with parameter sharing to reduce the number of parameters. The number of recursions is set to 4. The complete global spatial information extraction branch can be represented by (3).

C. Three-Dimensional Inception

Although the conventional multichannel 2-D convolution is convolved on 3-D data, the convolution kernel can only slide over the height and width of the feature map, while 3-D convolution is able to realize sliding on height, width, and depth dimensions. Excluding cases where the convolution kernel shape is $1 \times 1 \times 1$ and the stride is too large, such sliding usually involves elements repeatedly participating in the calculation. The process of sliding calculation is precisely the process of extracting and integrating the correlation of elements in the neighborhood. The sliding calculation in spectral dimension is also the advantage of 3-D convolution over 2-D convolution in extracting spectral information. However, the repeated participation of elements in the calculation will inevitably bring a high amount of computation.

In addition to the drawback of huge computational effort, conventional 3-D convolution faces another challenge. A convolution kernel of shape $k \times k \times k$ has k^3 parameters. When the network is trained, this set of parameters will have to learn the distribution of features in three dimensions. When a convolution kernel needs to perform linear fitting for multiple dimensions at the same time, the distribution learned in each dimension will be disturbed by information in other dimensions, resulting in parameter deviation. We tend to call this phenomenon "parameter contention" between different distributions. Since there is almost no set of parameters that can be used as the optimal solution for three complex distributions simultaneously, the parameter deviation seems to be unavoidable. This offset can easily lead to the learning of the feature distribution of spatial and spectral information is not accurate enough, and even counterproductive.

Scholars have put forward their own ideas on the use of 3-D convolution. Most of the work is to split the 3-D convolution, but this does not have a significant impact on the computational amount. To address the above two problems, our solution is to specialize the utility of 3-D convolution, that is, to simplify the role of 3-D convolution in the model and make its effects more specific. Specifically, our model is a three-branch network that extracts global spatial information, local spatial information, and spectral information, respectively. The proposed 3-D inception module is only responsible for the extraction of spectral information. As shown in Fig. 3, all 3-D convolution kernels in this module only need to preserve the scale of the spectral dimension, and the shape of the kernel is $k \times 1 \times 1$. It means that in this module, the repeated computation occurs only in the spectral dimension, which greatly reduces the computational cost. Inspired by GoogLeNet [54], we set different k to realize the extraction of multiscale spectral information, and the operation on each scale can be expressed as

$$I_k = f_k\left(I^0\right) \tag{13}$$

where f_k represents the three-layer 3-D convolution function with different k. In MSSR, we set the k to 3, 5, and 7. Finally, the information of different scales is connected on the channel dimension, that is,

$$I_{3\text{D}} = \text{Concat}\left(I_3, I_5, I_7, I^0\right). \tag{14}$$

It is worth noting that all of the above convolution layers are accompanied by activation functions afterward.

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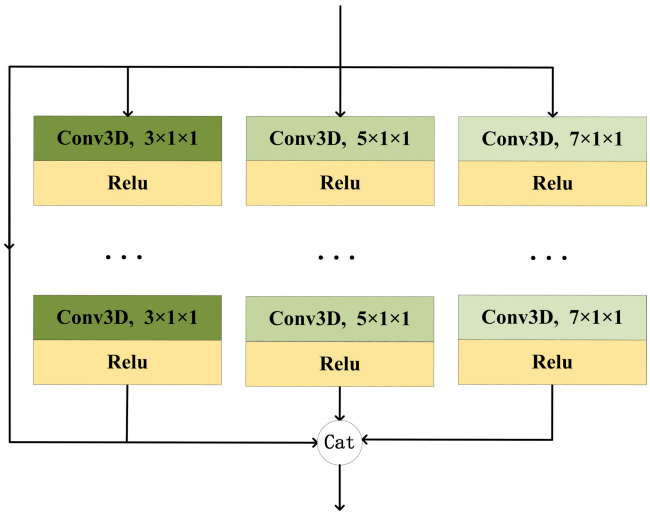


Fig. 3. Structure of the proposed 3-D inception.

IV. Experiments and Results

In this section, we will provide a comprehensive evaluation of MSSR. First, we will introduce four benchmark datasets and the details of the experiment setup. Then, MSSR will be quantitatively compared with other SOTA methods, including four objective image quality assessment metrics, the number of model parameters, and the amount of computation. Finally, we set up two sets of ablation experiments and a real scene SR task to verify the effectiveness of the proposed structure.

A. Datasets

1) MDAS: MDAS[1] is a multimodal benchmark dataset from the city of Augsburg, Germany, published by the Technical University of Munich (TUM) that can be used for SR tasks. MDAS includes synthetic aperture radar (SAR) data, MSIs, HSIs, digital surface model (DSM), and geographic information system (GIS) data. The observed scene of this dataset contains relatively complete urban elements, such as buildings, roads, rivers, woods, and so on, and the proportion of each type of element is relatively balanced. We chose subregion 1 as the target image for this experiment, which has a spatial size of 1364 × 1636 and 368 spectral bands in the range of 417–2484 nm. In order to reduce the cost, we intercept the first 120 bands for the experiment. The 500 × 500 region in the upper left corner of the image was used as the testing set.

2) Chikusei: The Chikusei dataset is an urban and agricultural image of the neighborhood of Ibaraki, Japan, captured by the Headwall Hyperspec-VNIR-C sensor. The image has a spatial size of 2517 × 2335 and 128 spectral bands in the range of 363–1018 nm. The 500 × 500 region in the upper left corner of the image was used as the testing set.

3) Pavia Centre: The Pavia Centre dataset is a city image of Pavia captured by the reflective optics system imaging spectrometer (ROSIS) sensor. The image contains elements such as water, houses, trees, etc. The image has a spatial size of 1096 × 715 and 102 spectral bands in the range of 430–860 nm. The 300 × 200 region in the upper left corner of the image was used as the testing set.

4) Houston: The GRSS-DFC-2013-Houston dataset was collected by the National Center for Airborne Laser Mapping at the University of Houston in June 2012, and was distributed for the 2013 IEEE Geoscience and Remote Sensing Society Data Fusion Contest [55]. The Houston dataset is an image of the University of Houston campus and the neighboring urban area. The image contains trees, water, roads, and other elements. The image has a spatial size of 349 × 1905 and 144 spectral bands in the range of 380–1050 nm. The 349 × 400 region in the left part of the image was used as the testing set.

For the training set, we randomly flip the patches horizontally and rotate them to varying degrees. The augmented patches are then bicubic downsampled to obtain the LR HSI, which is the input to the network.

[1]https://mediatum.ub.tum.de/1657312

B. Implementation Details

In MSSR, we set the recursion number of the DSSA module in the global spatial information extraction branch to 4, and the number of cascaded residual blocks in the local spatial information extraction branch to 6. For the construction of the loss function, we use a combination of the $L1$ loss, the SAM loss, and the gradient loss. In this experiment, we choose a weight allocation of $1 : 0.3 : 0.1$. During training, we use the Adam optimizer ($\beta_1 = 0.9$ and $\beta_2 = 0.999$), the batch size is set to 4, and the initial learning rate is set to 0.0001. The total number of training epochs is 40, and the learning rate decreased tenfold after 20 epochs. To compare the performance of each method more fairly, all the training and testing processes of this experiment are done in the same environment. Our method is implemented on the PyTorch framework with NVIDIA GeForce GTX 3090 GPU.

The size of the input patches is $C \times 32 \times 32$, with C representing the number of bands of images. The next experiments deal with scale factors of 2–4. PSNR, SSIM SAM, and ERGAS are used to evaluate the reconstructed images. In addition, we will compare and analyze the number of parameters and the computational effort of the models.

C. Comparisons With the SOTA Methods

In this section, our method will be compared with seven representative methods, namely, Bicubic, 3D-FCNN [31], GDRRN [32], SSPSR [33], MCNet [35], ERCSR [36], and GELIN [34]. Among them, 3D-FCNN and GDRRN are early representative methods, MCNet and ERCSR are cutting-edge models based on 3-D convolution, and SSPSR and GELIN are excellent works based on band grouping. This set of experiments was carried out on four remote sensing hyperspectral datasets, MDAS, Chikusei, Pavia Centre, and Houston, with scale factors of 2–4.

1) Results on the MDAS Dataset: Table I shows the objective metrics results of all the SR methods on the MDAS dataset with different scale factors. The results prove that our method is significantly ahead of others. Taking the 2 × SR as an example, the result of bicubic interpolation is the worst. The structure of 3D-FCNN and GDRRN is relatively simple, and the ability to extract information is limited, so their reconstruction effect is not outstanding. MCNet and ERCSR both deeply explore the potential of 3-D convolution to extract spectral information. Compared with the earlier models, the performance of the two is greatly improved, and the PSNR

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TABLE I
AVERAGE QUANTITATIVE COMPARISONS ON MDAS DATASET BY FACTORS 2–4

S	Model	PSNR↑	SSIM↑	SAM↓	ERGAS↓
2	Bicubic	32.348	0.8664	4.765	15.722
	3D-FCNN [31]	33.137	0.8941	4.473	14.317
	GDRRN [32]	33.034	0.8874	4.572	14.564
	SSPSR [33]	33.79	0.9101	4.166	13.169
	MCNet [35]	33.77	0.9097	4.271	12.959
	ERCSR [36]	33.788	0.9105	4.221	12.926
	GELIN [34]	33.921	0.9143	3.832	12.676
	MSSR (Ours)	**34.322**	**0.9215**	**3.325**	**12.295**
3	Bicubic	30.308	0.788	5.968	13.321
	3D-FCNN [31]	30.872	0.8208	5.742	12.415
	GDRRN [32]	30.866	0.82	5.753	12.4
	SSPSR [33]	-	-	-	-
	MCNet [35]	31.135	0.8313	5.588	11.935
	ERCSR [36]	31.159	0.8328	5.569	11.9
	GELIN [34]	31.234	0.8366	5.151	11.775
	MSSR (Ours)	**31.451**	**0.8484**	**4.476**	**11.522**
4	Bicubic	29.007	0.712	7.19	11.501
	3D-FCNN [31]	29.389	0.7452	6.918	10.933
	GDRRN [32]	29.277	0.7221	6.986	11.101
	SSPSR [33]	29.722	0.7663	6.307	10.511
	MCNet [35]	29.604	0.7556	6.832	10.623
	ERCSR [36]	29.602	0.7574	6.887	10.619
	GELIN [34]	29.643	0.761	6.41	10.541
	MSSR (Ours)	**29.808**	**0.7745**	**5.6**	**10.383**

of the reconstructed images reaches 33.77 dB (−0.552 dB) and 33.788 dB (−0.534 dB), respectively. The SSPSR divides the LR HSI into multiple subimages from spectral dimensions for separate processing, which reduces the difficulty of information extraction while also losing some spectral correlation information to a certain extent. The method achieves slightly better results than ERCSR. On this basis, GELIN can be regarded as an improved version of SSPSR, and its proposed NGIM makes the spectral information well supplemented. The model achieves suboptimal results in both 2× and 3 × SR, with a PSNR of 33.921 dB (−0.401 dB) in the 2 × SR. Our method produces the best results in all cases, where the PSNR reaches 34.322 dB in the 2 × SR. Furthermore, the images reconstructed by MSSR have a far superior improvement in SAM, regardless of the scale factor. It is noteworthy that SSPSR, a relatively early model, achieves suboptimal results in the 4 × SR. This may be due to the fact that SSPSR employs the progressive upsampling framework that decomposes a difficult task into multiple relatively simple tasks, allowing the network to better cope with larger amplification factors. Although we simplify the structure of 3-D convolution, the experimental results demonstrate that retaining only the elements of the convolution kernel in spectral dimension can still accomplish the extraction of spectral correlation well. The proposed 3-D inception has a great advantage in improving spectral fidelity. Meanwhile, the attention module DSSA can help the network integrate spatial information better, and make the reconstructed image have good spatial quality.

Fig. 4 shows the visualization results of the reconstructed images in the 4 × SR as well as the error heat map. Band 10 (450 nm), band 40 (558.67 nm), and band 71 (670.96 nm) were selected for false-color image synthesis. Overall, the results of Bicubic, 3D-FCNN, and GDRRN are not satisfactory, while the reconstructed images produced by MSSR possess the best visual effects. In terms of color fidelity, the reconstructed images of MCNet, ERCSR, and GELIN show more obvious color distortion. The results of SSPSR and MSSR are the best to restore the labeled HR images, and the reconstructed images of MSSR have better color saturation compared to SSPSR. This is because our proposed 3-D inception module can capture multiscale spectral information, which can provide more information support for spectral reconstruction. From the perspective of detailed texture, our method has excellent performance in capturing spatial information, and the recovered houses and trees have clearer and cleaner edge contours without excessive smoothing. Taking the area marked by the red box as an example, we can only recognize the observed object as a house from the image reconstructed by MSSR, and it also shows lower heat in the heat map. This also proves that the global and local spatial information extraction branches in MSSR can make up for the problem of inadequate spatial information extraction that exists in other methods.

TABLE II
COMPARISON OF PARAMETER QUANTITY AND COMPUTATION AMOUNT

Model	Parameters(M)	GFLOPs
SSPSR [33]	11.1	90.29
MCNet [35]	1.9	236.61
ERCSR [36]	1.3	178.84
GELIN [34]	26.7	563.65
MSSR (Ours)	24.5	49.64

To more intuitively show the remarkable performance of our method in improving spectral fidelity, Fig. 5 provides the mean spectral difference curves of the reconstructed and labeled images in the 2 × SR. It is obvious that the reconstructed spectrum of MSSR has the least difference from the labeled spectrum and can recover the spectral features of the target well. Other methods exhibit more severe spectral distortion.

Table II shows the number of parameters and computational effort of each model in recent years for the 2 × SR on the MDAS dataset. GELIN and MSSR have the most parameter requirements, while MCNet and ERCSR require only a very low number of parameters. The parameters of our method mainly come from the two spatial information extraction branches, and the number of parameters of the multiscale spectral information extraction branch is only 2.1 M. MCNet and ERCSR are 3-D convolution-based models. As we know from our previous work [56], 3-D convolution usually has a smaller number of parameters and higher computational cost compared to multichannel 2-D convolution, and this characteristic is also reflected in these two methods. Compared with other models, the computational effort of our method is greatly reduced. Specifically, SSPSR, MCNet, ERCSR, and GELIN are 1.82, 4.77, 3.6, and 11.35 times more computationally expensive than our method, respectively. Such low computational cost is mainly due to our special utilization of 3-D convolution, avoiding repeated computations in spatial dimensions greatly reduces the computational requirement of the 3-D inception module in MSSR.

2) Results on Chikusei Dataset: Table III shows the objective metrics results of all the SR methods on the Chikusei dataset with different scale factors. In terms of the overall numerical performance, the trends presented are basically consistent with the MDAS dataset. Taking 2 × SR as an example,

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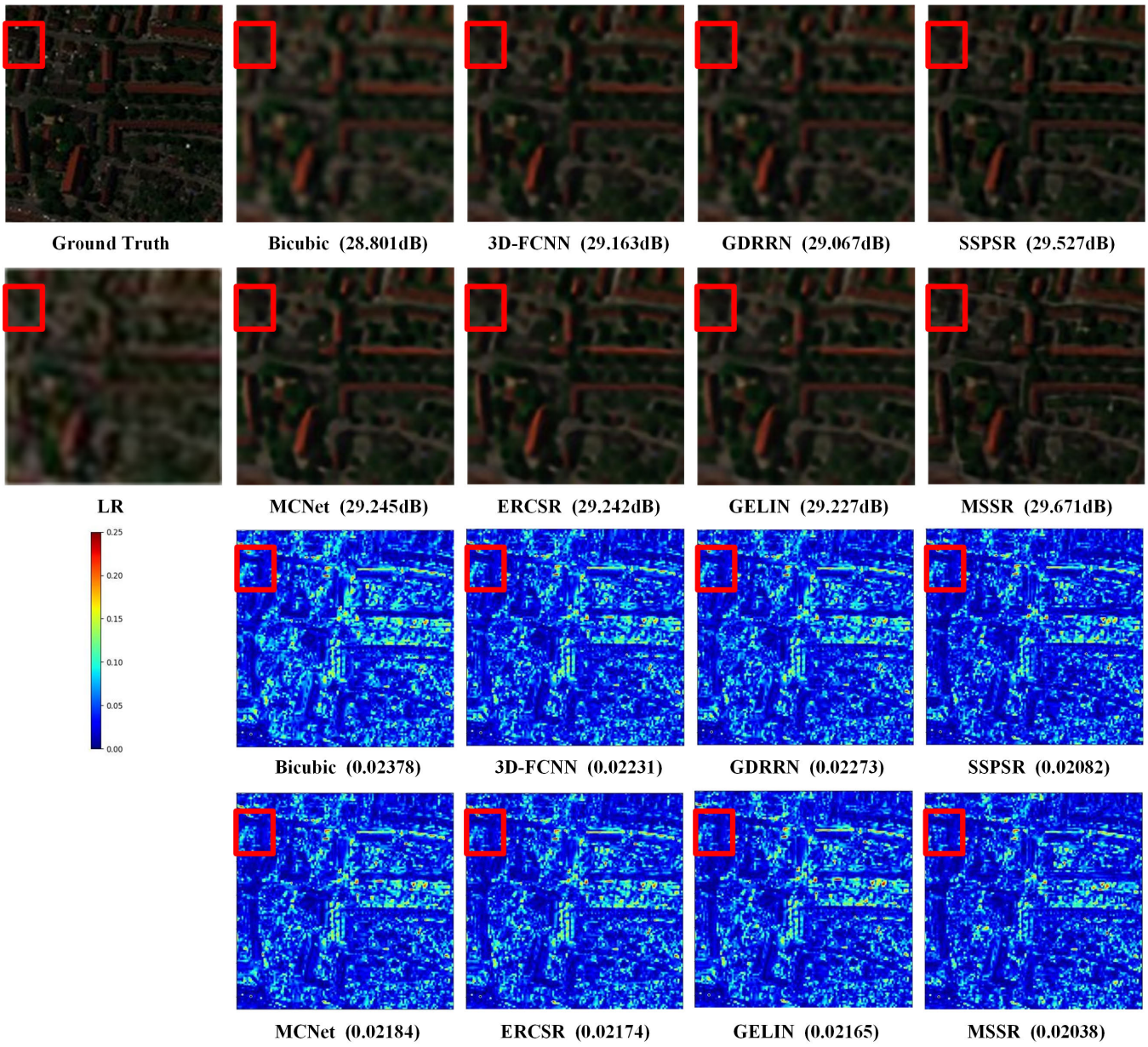


Fig. 4. Reconstructed images and error heat maps of one test HSI from the MDAS dataset for the 4 × SR.

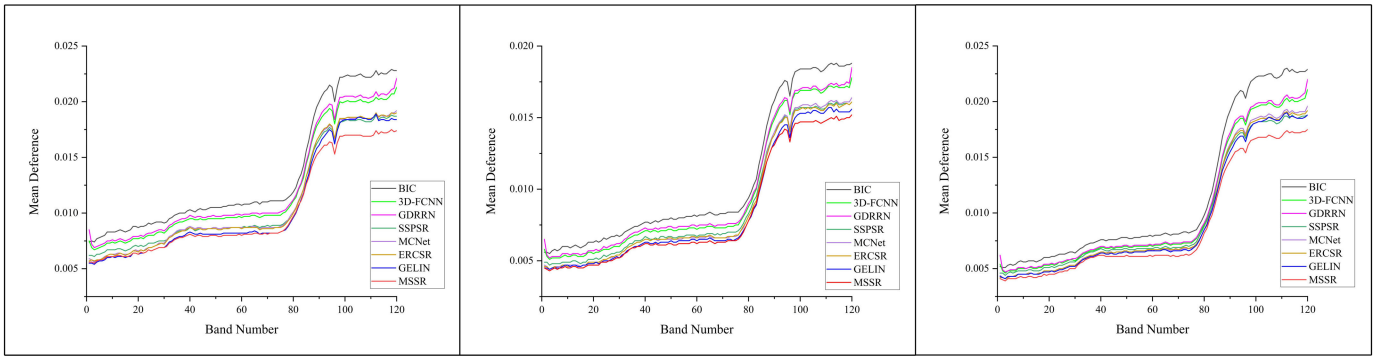


Fig. 5. Mean spectral difference curve of three test HSIs from the MDAS dataset for the 2 × SR.

the PSNR, SSIM, SAM, and ERGAS of the reconstructed results of our method are 44.714 dB, 0.9845, 1.706, and 5.618 (−0.193), respectively. The performance of GELIN of the four metrics is 44.422 dB (−0.292 dB), 0.9835 (−0.001), 1.738 (−0.032), and 5.425, respectively. The results of MCNet and ERCSR are slightly worse than those of GELIN and MSSR but still achieve an obvious lead relative to other models. It can be seen that the special processing done by scholars for the

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TABLE III
AVERAGE QUANTITATIVE COMPARISONS ON CHIKUSEI DATASET BY FACTORS 2–4

S	Model	PSNR↑	SSIM↑	SAM↓	ERGAS↓
2	Bicubic	41.247	0.9668	2.262	7.963
	3D-FCNN [31]	42.881	0.9778	1.942	6.590
	GDRRN [32]	42.56	0.9732	2.125	6.64
	SSPSR [33]	43.526	0.9806	2.084	6.669
	MCNet [35]	44.41	0.9834	1.789	5.582
	ERCSR [36]	44.349	0.9833	1.798	5.513
	GELIN [34]	44.422	0.9835	1.738	**5.425**
	MSSR (Ours)	**44.714**	**0.9845**	**1.706**	5.618
3	Bicubic	37.318	0.919	3.376	8.469
	3D-FCNN [31]	38.274	0.9359	3.097	7.562
	GDRRN [32]	38.155	0.9291	3.154	7.751
	SSPSR [33]	-	-	-	-
	MCNet [35]	39.257	0.9473	2.82	6.593
	ERCSR [36]	39.265	0.9474	2.795	6.558
	GELIN [34]	39.317	0.9481	2.696	6.523
	MSSR (Ours)	**39.753**	**0.953**	**2.482**	**6.484**
4	Bicubic	35.338	0.8781	4.133	8.071
	3D-FCNN [31]	35.989	0.8964	3.843	7.482
	GDRRN [32]	35.886	0.8878	3.914	7.532
	SSPSR [33]	37.017	0.916	3.337	6.708
	MCNet [35]	36.818	0.9117	3.49	6.654
	ERCSR [36]	36.837	0.9118	3.487	6.624
	GELIN [34]	36.878	0.913	3.385	6.548
	MSSR (Ours)	**37.294**	**0.9206**	**3.113**	**6.462**

TABLE IV
AVERAGE QUANTITATIVE COMPARISONS ON PAVIA CENTREDATASET BY FACTORS 2–4

S	Model	PSNR↑	SSIM↑	SAM↓	ERGAS↓
2	Bicubic	32.539	0.9085	4.615	7.820
	3D-FCNN [31]	34.251	0.937	4.075	6.536
	GDRRN [32]	33.755	0.924	4.258	6.944
	SSPSR [33]	35.314	0.9475	3.983	5.884
	MCNet [35]	35.587	0.9506	3.749	5.656
	ERCSR [36]	35.612	0.9509	3.726	5.635
	GELIN [34]	35.647	0.9513	3.668	5.614
	MSSR (Ours)	**36.236**	**0.9552**	**3.629**	**5.373**
3	Bicubic	29.223	0.8062	5.692	7.385
	3D-FCNN [31]	30.41	0.8554	5.223	6.518
	GDRRN [32]	30.121	0.8365	5.463	6.853
	SSPSR [33]	-	-	-	-
	MCNet [35]	31.165	0.8753	5.064	5.999
	ERCSR [36]	31.168	0.8754	5.057	5.986
	GELIN [34]	31.16	0.8753	4.917	6.002
	MSSR (Ours)	**31.547**	**0.8836**	**4.768**	**5.792**
4	Bicubic	27.284	0.7053	6.313	6.791
	3D-FCNN [31]	28.08	0.758	6.082	6.242
	GDRRN [32]	27.955	0.741	6.153	6.212
	SSPSR [33]	**28.845**	**0.7968**	5.681	**5.739**
	MCNet [35]	28.649	0.7867	6.018	5.864
	ERCSR [36]	28.644	0.7869	6.073	5.86
	GELIN [34]	28.63	0.7848	5.814	5.878
	MSSR (Ours)	28.724	0.7881	**5.678**	5.843

3-D property of HSIs, either the modifying of 3-D convolution or the band grouping strategy, can achieve impressive results. For 3× and 4 × SR, our method achieves varying degrees of lead in all four metrics. GELIN is suboptimal for 2× and 3 × SR, while SSPSR still achieved suboptimal results for 4 × SR. This shows that the band grouping strategy can provide a good inspiration for the HSI SR task. However, two basic difficulties exist in this direction. One is that when spectral bands are divided, it will inevitably cause the loss of spectral correlations. How to effectively supplement this kind of information is a problem worthy of further investigation. Second, how to alleviate the information redundancy when reintegrating subimages is also a major difficulty. The numerical results on this dataset also prove the effectiveness of our proposed multidomain spatial information and multiscale spectral information extraction strategy.

Fig. 6 shows the visualization results of the reconstructed images in the 4× SR as well as the error heat map. We select bands 30, 50, and 70 for false-color image synthesis. In terms of color fidelity, each method has a different degree of color distortion, but our results are still the closest to the labeled image. Take the area marked by the red box as an example. By observing the outline of the white building, we can find that the image reconstructed by MSSR has the clearest edge, and the spatial size is closest to the labeled image. On the contrary, the reconstruction results of other methods have a more obvious blurring effect on the boundaries of the white building, and the spatial size is much larger than the labeled image.

Fig. 7 provides the mean spectral difference curve between the reconstructed image and the labeled image of each model for 2 × SR. It can be clearly observed that MSSR can reconstruct the most realistic spectral information. In addition, the method based on 3-D convolution has an outstanding ability to improve spectral fidelity.

3) Results on Pavia Centre Dataset: Table IV shows the objective metrics results of all the SR methods on the Pavia Centre dataset with different scale factors. The MSSR achieves the best results in all four metrics in both 2× and 3 × SR and significantly improved compared with the suboptimal results. In the 2 × SR task, GELIN has suboptimal results in all metrics. MCNet, ERCSR, and GELIN have similar performance in 3 × SR tasks, all performing well. Unlike the previous situation, the PSNR, SSIM, and ERGAS of reconstructed images of MSSR are only suboptimal in the 4 × SR task, but it still perform best in spectral fidelity, which also confirms the powerful capability of the 3-D inception module. SSPSR is still strong when dealing with large-scale factors. From this result, it can be seen that the progressive upsampling framework has a relatively stable and excellent performance in handling large-scale SR tasks.

Fig. 8 shows the visualization results of the reconstructed images in the 4 × SR as well as the error heat map. We select bands 30, 50, and 70 for false-color image synthesis. Surprisingly, although our method is not optimal in metrics, it is the best performer in visual effects. In terms of color fidelity, the reconstruction result of MSSR is the best, SSPSR and ERCSR are slightly worse, and the results of other methods have severe color distortion. In terms of details, the advantage of our method is more obvious. The recovery of the contours of the houses, the fine textures of the roofs, and the shadows of the houses on the street by MSSR are the closest to the ground truth. Even the roofs reconstructed by SSPSR lose most of the texture information. Similar to the performance of the other two datasets, our method performs best in the mean spectral difference curve shown in Fig. 9.

4) Results on Houston Dataset: Table V shows the objective metrics results of all the SR methods on the Houston

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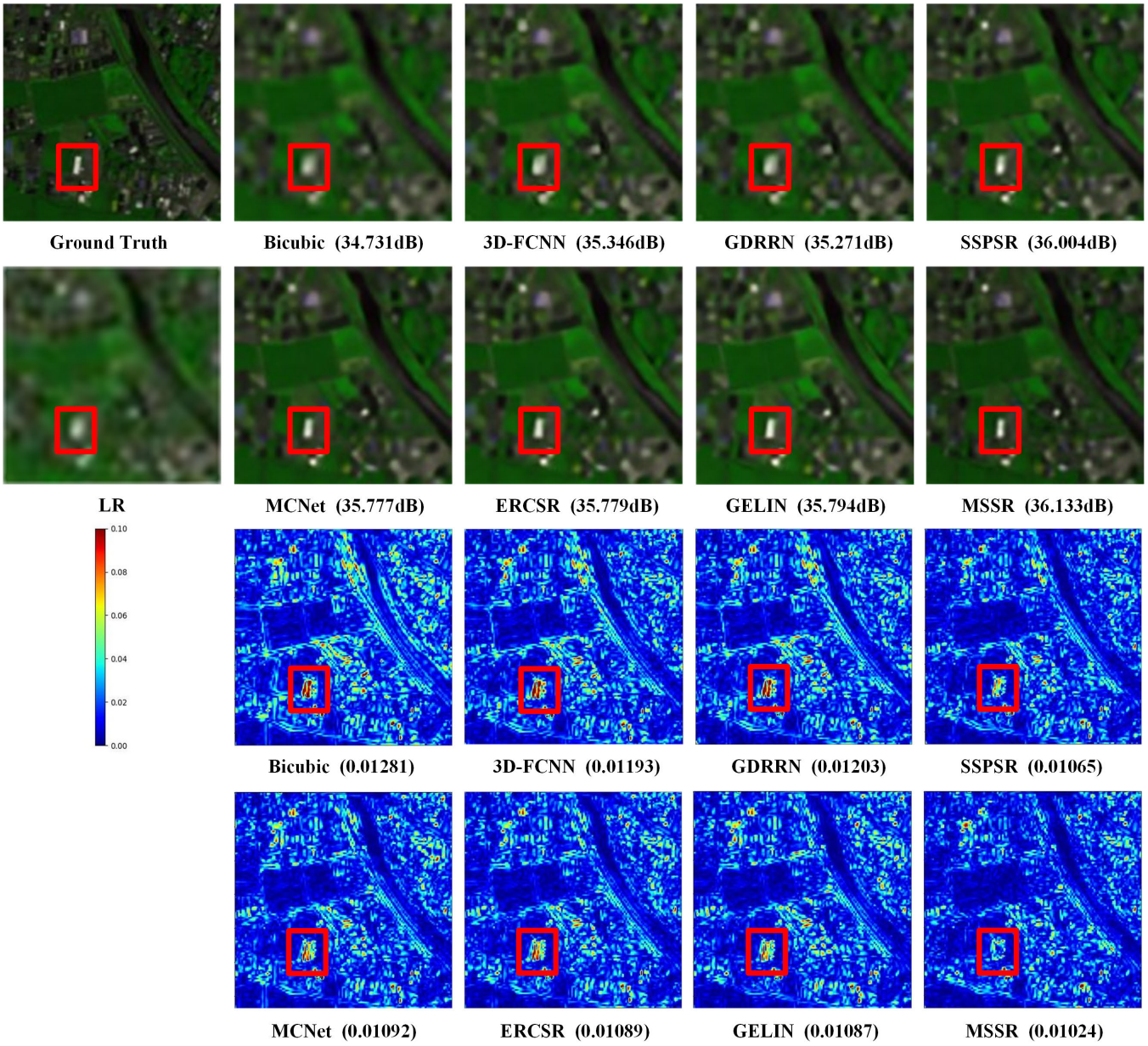


Fig. 6. Reconstructed images and error heat maps of one test HSI from the Chikusei dataset for the 4 × SR.

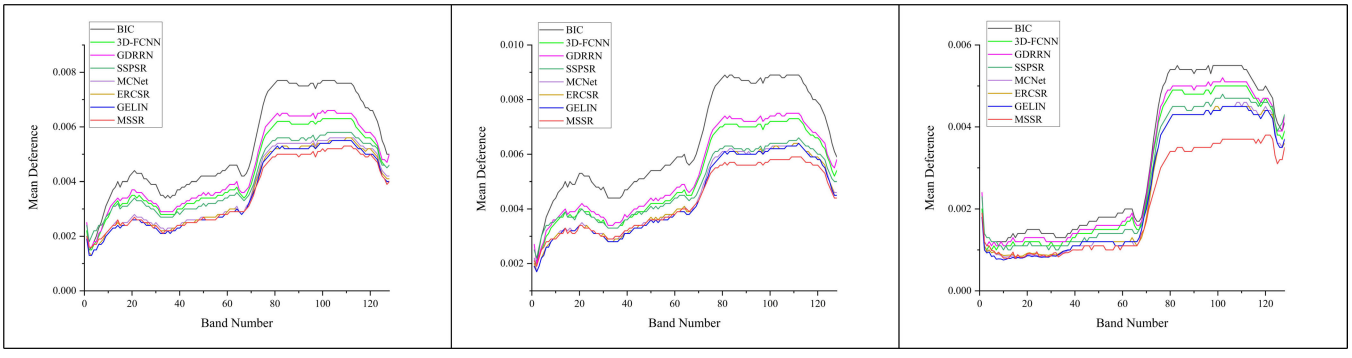


Fig. 7. Mean spectral difference curve of three test HSIs from the Chikusei dataset for the 2 × SR.

dataset with different scale factors. Overall, our method achieves the best results on all three scales. Suboptimal results for 2× and 3 × SR were provided by GELIN, while SSPSR performed well at 4 × SR. The SAM provided by our method has a large degree of lead. Under the three scales, the results are improved by 0.341 (11.7%), 0.368 (9%), and 0.159 (3.2%), respectively, compared with the suboptimal results. This proves the excellent performance

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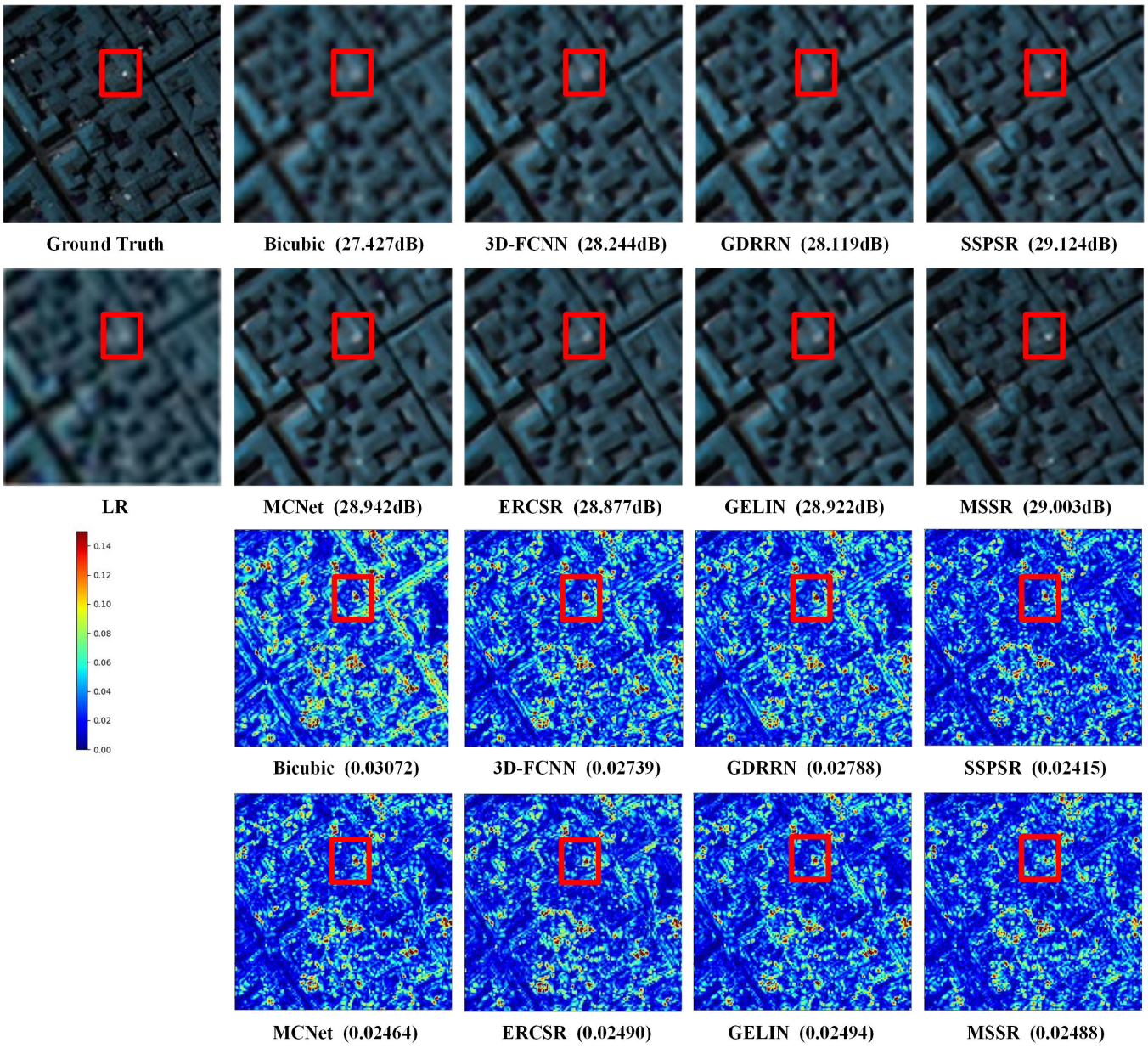


Fig. 8. Reconstructed images and error heat maps of one test HSI from the Pavia Centre dataset for the 4 × SR.

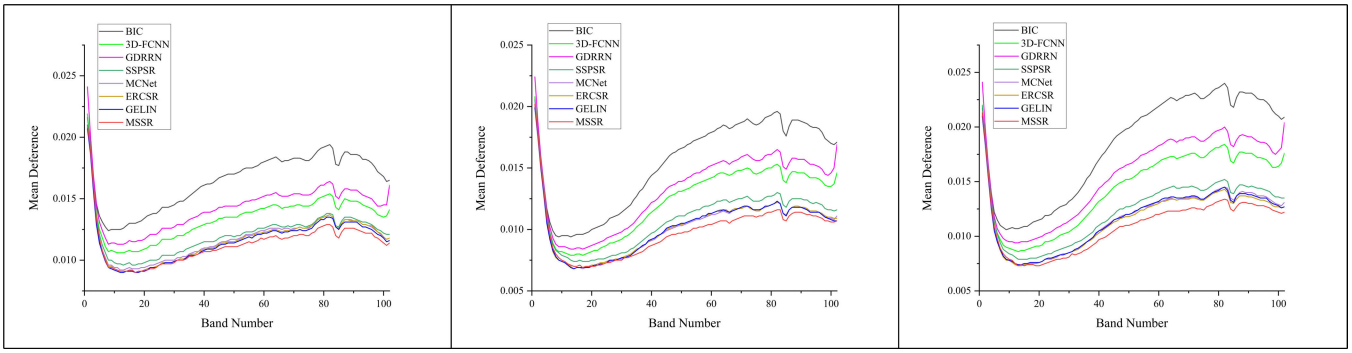


Fig. 9. Mean spectral difference curve of three test HSIs from Pavia Centre dataset for the 2 × SR.

of our proposed multiscale spectral feature extraction module.

Fig. 10 shows the visualization results of the reconstructed images in the 4 × SR as well as the error heat map. We select bands 20, 40, and 60 for false-color image synthesis. Observing the upper right part of the image, it can be found that the image reconstructed by our method has a significantly richer texture. In addition, the road reconstructed by MSSR is

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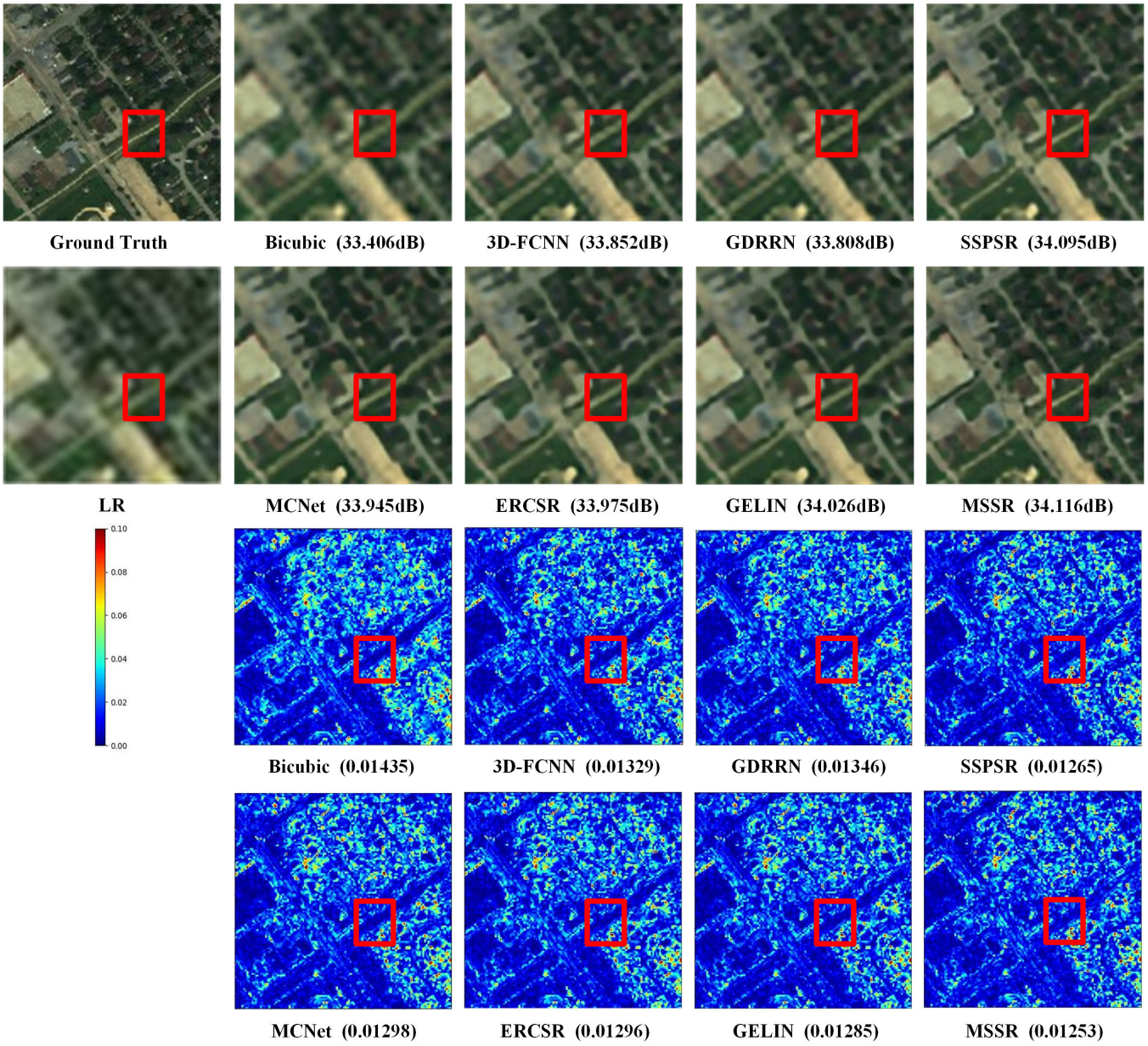


Fig. 10. Reconstructed images and error heat maps of one test HSI from the Houston dataset for the 4 × SR.

slenderer and has a less fuzzy effect than other methods. In the mean spectral difference curve shown in Fig. 11, our method also has obvious advantages.

D. Ablation Study

In this section, we will conduct a comprehensive experimental verification of the effectiveness of the proposed method. On the one hand, we will verify the effectiveness of the DSSA and 3-D inception modules in the network, and on the other hand, we will compare the performance of the special 3-D convolutional structure within the 3-D inception module with regular multichannel 2-D convolution and 3-D convolution. All experiments in this section are based on the MDAS dataset and deal with the 2 × SR task.

1) Effectiveness of DSSA and 3-D Inception: To validate the effectiveness of our proposed two branches, DSSA and 3-D inception, we designed this set of comparison experiments. Specifically, this set of experiments will examine the performance of four types of networks, namely, the complete MSSR model, the incomplete MSSR without the DSSA or 3-D inception module, and the network that only retains the local spatial information extraction branch.

We compare the PSNR, SSIM, SAM, and ERGAS of the reconstructed images of the four networks, and Table VI shows all the experimental results. Overall, the complete MSSR model performs best in all four metrics. By observing SAM, it can be found that the two networks using the 3-D inception module obtain optimal and suboptimal results, respectively, which indicates that the ability of the 3-D inception module to mine spectral information is quite outstanding. Compared with "MSSR w/o DSSA and 3-D inception," which only uses the local spatial information extraction branch, "MSSR w/o 3-D inception," which removes the spectral information extraction

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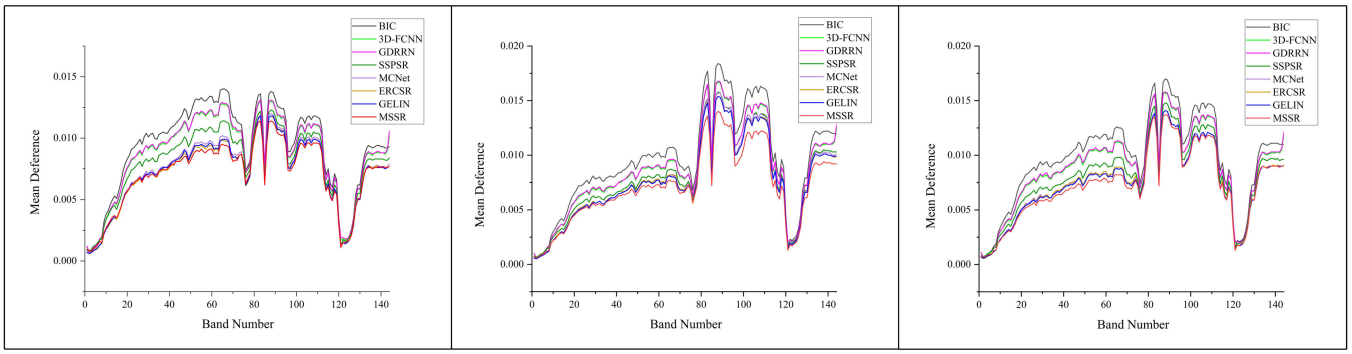


Fig. 11. Mean spectral difference curve of three test HSIs from the Houston dataset for the 2 × SR.

TABLE V
AVERAGE QUANTITATIVE COMPARISONS ON HOUSTON DATASET BY FACTORS 2–4

S	Model	PSNR↑	SSIM↑	SAM↓	ERGAS↓
2	Bicubic	35.854	0.9149	3.68	7.826
	3D-FCNN [31]	36.594	0.9303	3.391	7.235
	GDRRN [32]	36.532	0.9296	3.427	7.3
	SSPSR [33]	37.311	0.9396	3.14	6.647
	MCNet [35]	37.72	0.9437	2.98	6.284
	ERCSR [36]	37.724	0.9436	2.966	6.28
	GELIN [34]	37.764	0.9444	2.913	6.253
	MSSR (Ours)	**38.087**	**0.949**	**2.572**	**6.058**
3	Bicubic	33.388	0.8561	4.873	6.624
	3D-FCNN [31]	34.12	0.8785	4.448	6.118
	GDRRN [32]	33.961	0.8762	4.523	6.235
	SSPSR [33]	-	-	-	-
	MCNet [35]	34.718	0.8903	4.165	5.668
	ERCSR [36]	34.766	0.8913	4.137	5.633
	GELIN [34]	34.785	0.892	4.067	5.631
	MSSR (Ours)	**34.938**	**0.8971**	**3.699**	**5.544**
4	Bicubic	31.931	0.8026	6.169	5.907
	3D-FCNN [31]	32.587	0.8283	5.618	5.485
	GDRRN [32]	32.489	0.8267	5.677	5.552
	SSPSR [33]	33.124	0.8460	4.967	5.15
	MCNet [35]	32.977	0.8395	5.374	5.218
	ERCSR [36]	32.996	0.8399	5.367	5.206
	GELIN [34]	33.04	0.8413	5.246	5.186
	MSSR (Ours)	**33.145**	**0.8468**	**4.808**	**5.133**

branch, does not improve the metrics significantly, while the "MSSR w/o DSSA" with the global spatial information extraction branch removed has a great improvement in four metrics. When DSSA and 3-D inception work together, the performance of reconstructed images can be further improved, which conveys the message to us that when designing a network, the orthogonality of modules should be considered along with the synergy between modules. According to the trend of the above numerical results, it is not difficult to conclude that both the proposed DSSA module, which is responsible for global spatial information extraction, and the 3-D inception module, which is responsible for multiscale spectral information extraction, have captured image information very effectively and achieved good reconstruction effects.

2) Study of 3-D Inception: The purpose of this set of experiments is to compare the performance differences between multichannel 2-D convolution, conventional 3-D convolution, and 3-D convolution that only preserves the spectral dimension elements used in the 3-D inception module. In addition to the four reconstructed image quality evaluation metrics, we will also compare the number of parameters and computational effort of each model. Specifically, we replace the convolution in the 3-D inception module with multichannel 2-D convolution or regular 3-D convolution, which are named MSSR_2D and MSSR_3D, respectively.

Table VII shows the experimental results of each model. The PSNR and SAM of the reconstructed image of MSSR are both optimal, and the SSIM is only 0.0003 lower than the best result. MSSR_3D also has outstanding performance in metrics. In addition to providing the highest SSIM and ERGAS, the PSNR and SAM of its reconstructed images are also suboptimal, and have a significant lead relative to MSSR_2D. MSSR_2D is lagging behind on all fronts. We are more interested in the effect of different convolutional modes on the number of parameters and the amount of computation than the image quality metrics. Comparing MSSR_2D and MSSR_3D first, the results are consistent with the general rule that 3-D convolution typically has a smaller number of parameters and higher computational cost relative to 2-D convolution. As expected, the 3-D convolution that retains only the spectral dimension elements achieves a smaller number of parameters than the regular 3-D convolution and a lower computational effort than the multichannel 2-D convolution. The reason for this is that only the elements of the 3-D convolution kernel in the spectral dimension are preserved with a $k \times 1 \times 1$ kernel shape. It naturally requires only a smaller number of parameters compared to the conventional 3-D convolution with a $k \times k \times k$ kernel shape. This particular convolution kernel shape, when performing convolution operations, will only cause the feature map elements to participate in the calculation repeatedly in the spectral dimension due to sliding. In contrast, multichannel 2-D convolution and conventional 3-D convolution will have a large number of repeated computations in two and three dimensions, respectively. This property is also reflected in the GFLOPs of the three models. Reviewing the overall performance of these three convolutional modes, the first thing that can be determined is that 3-D convolution is indeed better than 2-D convolution in mining spectral information. Second, our proposed 3-D inception module performs well in both computational cost and spectral information extraction. Focusing on future development, the high amount of computation brought by 3-D convolution will bring a great burden to the model, and our method provides a valuable research idea to realize the efficient utilization of 3-D convolution.

E. SR for Real HSIs

In order to evaluate the generalization and application potential of our method, we carry out SR reconstruction

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TABLE VI

EFFECTIVENESS OF DSSA AND 3-D INCEPTION. AVERAGE QUANTITATIVE COMPARISONS AMONG SOME VARIANTS OF THE PROPOSED METHOD ON MDAS DATASET BY FACTOR 2

Model	PSNR↑	SSIM↑	SAM↓	ERGAS↓
MSSR	**34.322**	**0.9215**	**3.325**	**12.295**
MSSR w/o DSSA	34.258	0.9203	3.382	12.445
MSSR w/o 3D Inception	34.203	0.9176	3.526	12.695
MSSR w/o DSSA and 3D Inception	34.114	0.9166	3.491	12.657

TABLE VII

STUDIES OF 3-D INCEPTION. AVERAGE QUANTITATIVE COMPARISONS AMONG SOME VARIANTS OF THE 3-D INCEPTION ON MDAS DATASET BY FACTOR 2

Model	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Parameters (M)	GFLOPs
MSSR	**34.322**	0.9215	**3.325**	12.295	**24.5**	**49.64**
MSSR _2D	34.192	0.919	3.426	12.488	46.5	68.04
MSSR _3D	34.319	**0.9218**	3.332	**12.268**	27.7	84.88

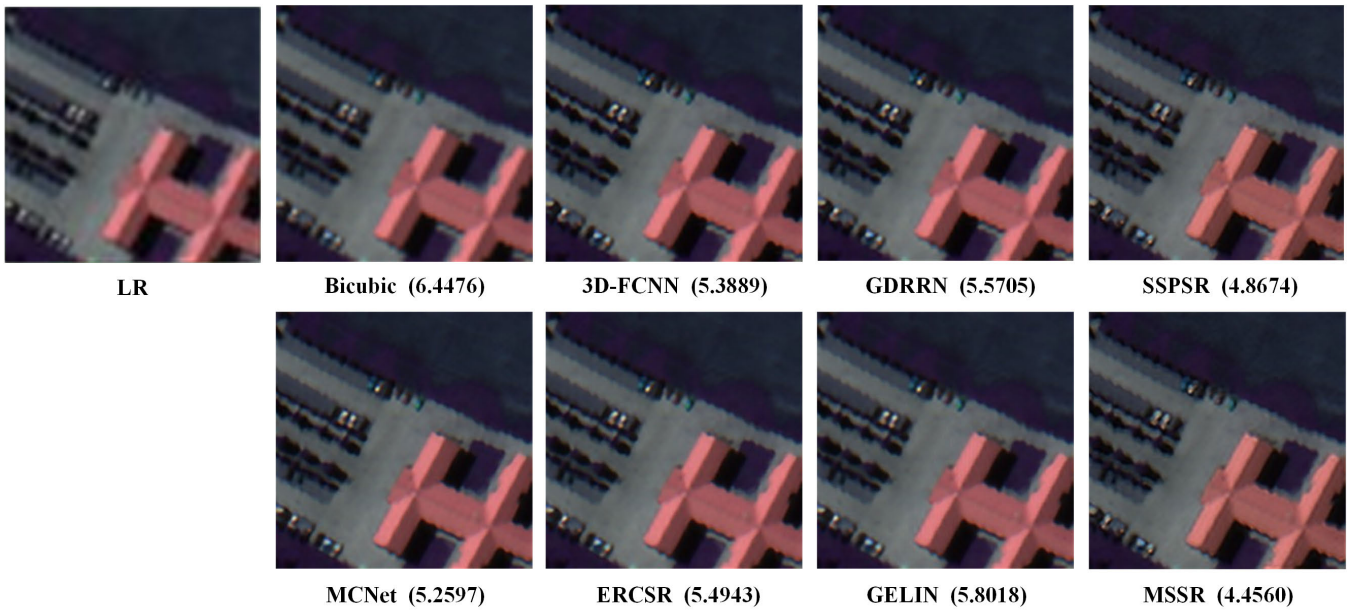


Fig. 12. Reconstructed images of real scenes from the Pavia University dataset for the 4 × SR.

for real scenes in the Pavia University dataset. We test the model trained by LR–HR image pairs in the Pavia Centre dataset. Both datasets were captured by ROSIS sensor, with 103 bands in Pavia University and 102 bands in Pavia Centre. We test the first 102 bands in the Pavia University dataset. In addition, in order to make an objective assessment of the quality of reconstructed images without reference images, we introduce the natural image quality evaluator (NIQE) index.

Fig. 12 shows the visualization results of each model for 4 × SR. We annotate their NIQE values below each image. Both qualitative and quantitative results demonstrate the remarkable performance of our method. This means that our method has great generalization ability and can deal well with SR tasks in real scenes.

V. CONCLUSION

In this article, we propose a new SR reconstruction method MSSR for remote sensing HSIs. The model effectively improves the ability of the network to extract multidomain spatial information and multiscale spectral information and largely reduces the computational burden of the network while realizing high-quality reconstruction of remote sensing HSIs. Specifically, we design a three-branch network for global spatial information extraction, local spatial information extraction, and multiscale spectral information extraction, respectively. Inspired by image self-similarity, we propose a spatial–spectral attention mechanism based on dilated convolution for global spatial information extraction, which effectively solves the problem of inadequate spatial information extraction. In addition, we design a 3-D inception module for multiscale spectral information extraction. By only retaining the elements of the 3-D convolution kernel in the spectral dimension, the elements in the feature map only participate in the calculation repeatedly in one dimension, which greatly reduces the computational cost while realizing the effective extraction of spectral information. Complete ablation experiments demonstrate the effectiveness of our proposed module and method. Compared with the advanced methods, our method not only achieves SOTA results on the four image evaluation metrics but also greatly reduces the calculation cost of the model. Meanwhile, from the visualization results, the images reconstructed by our method have more realistic spectral information and better-detailed texture.

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